

The Neuron

Note: In OneNote, set this worksheet as background. Use "Export" in the simulation and insert the image in the marked space.

Neurons receive, integrate, conduct and transmit information. Dendrites receive postsynaptic signals; the soma integrates them and controls metabolism. At the axon hillock, excitatory and inhibitory influences are summed. If the threshold is exceeded, an action potential is triggered. Voltage-gated ion channels conduct it along the axon. In myelinated fibres, excitation jumps from node to node, which increases speed and saves energy. At the terminal buttons, the electrical signal becomes chemical: calcium influx triggers vesicle fusion, neurotransmitters enter the synaptic cleft and bind to receptors on the postsynaptic membrane.

Task 1: Label the neuron

Export the labelled figure from the "Neuron" mode and insert it below.

Insert here

Task 2: Matching structure and function

Complete the matching task, export the correct solution and insert it below.

Insert here

Task 3: Signal transmission along the neuron

Describe the signal path from dendrites through soma and axon to the terminal buttons. Use technical terms.

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.